

Same graph, different samples, different apparent persistence

ρ_2 = share of pairs active in ≥ 2 windows

Full network

pair 1 2 3 4 5 count

A ● ● ● ● ● 1

B ● — ● — — 1

C — — ● ● 1

D ● — — — 0

E — ● — — 0

F — — — ● 0

3/6 = 50 %

true persistence

R · random nodes

1 2 3 4 5 count

A — — — — —

B ● — ● — — 1

C — — ● ● 1

D ● — — — 0

E — ● — — 0

F — — — — —

2/4 = 50 %

sample: same share

S · random walk

1 2 3 4 5 count

A ● ● ● ● ● 1

B ● — ● — — 1

C — — ● ● 1

D — — — — —

E — ● — — 0

F — — — — —

3/4 = 75 %

sample: +25 pp

H · late time only

1 2 3 4 5 count

A — — — — —

B — — ● — — 0

C — — ● ● 1

D — — — — —

E — ● — — 0

F — — — — —

1/3 = 33 %

sample: -17 pp

B · event loss

1 2 3 4 5 count

A ● — ● — ● 1

B — — ● — — 0

C — — — — —

D — — — — —

E — ● — — 0

F — — — — —

1/3 = 33 %

sample: -17 pp